

AI-Powered Virtual Meeting Summarization System

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Abstract—Virtual meetings are a vital component of modern remote and hybrid work environments, yet reviewing complete recordings or transcripts is often inefficient and time-consuming. This research presents an AI-driven Virtual Meeting Summarization System that automates transcription, analysis, and summarization of meeting content using advanced Natural Language Processing (NLP) and Machine Learning (ML) techniques. The system integrates Automatic Speech Recognition (ASR), speaker diarization, and Google’s Gemini-based summarization framework to capture key discussion points and accurately attribute speaker contributions. The system’s performance is validated using Word Error Rate (WER) for transcription and ROUGE metrics for summarization, demonstrating high precision and operational efficiency. Comparative analysis with existing summarization systems shows that our approach significantly reduces manual documentation efforts while enhancing knowledge retention and collaborative decision-making in virtual settings.

Index Terms—Virtual Meetings, Natural Language Processing, Machine Learning, Meeting Summarization, Collaboration

I. INTRODUCTION

In today’s rapidly evolving remote and hybrid work culture, virtual meetings have become a foundational element of communication across industries, academia, and organizations. Platforms such as Zoom, Microsoft Teams, and Google Meet have enabled seamless connectivity, breaking geographical barriers. However, while these tools have improved accessibility, they have also introduced a persistent challenge—documenting, summarizing, and retrieving critical meeting information efficiently.

Traditionally, meeting documentation has relied on manual note-taking or post-meeting transcript reviews, both of which are time-consuming and prone to human error. Over the past decade, significant advancements in Natural Language Processing (NLP) and Machine Learning (ML) have led to the development of automated summarization systems aimed at reducing this burden. Early systems, such as extractive summarizers like TextRank, focused on selecting the most important sentences based on word frequency and graph-based algorithms. While effective to some extent, these approaches often lacked semantic understanding and context retention, especially in dynamic, multi-speaker meetings.

The emergence of large language models (LLMs) such as GPT-4, PaLM-2, and Gemini has revolutionized the summarization landscape by enabling abstractive summarization—capable of generating new sentences that concisely represent the essence of discussions. Yet, despite these advancements, existing solutions frequently fall short in capturing

speaker-specific contributions, handling overlapping dialogues, and automating

The remainder of this paper is organized as follows: Section II discusses related work; Section III outlines the methodology and system architecture; Section IV details implementation results; Section V presents evaluation and discussion; Section VI highlights challenges and limitations; Section VII proposes future enhancements; and Section VIII concludes the paper.

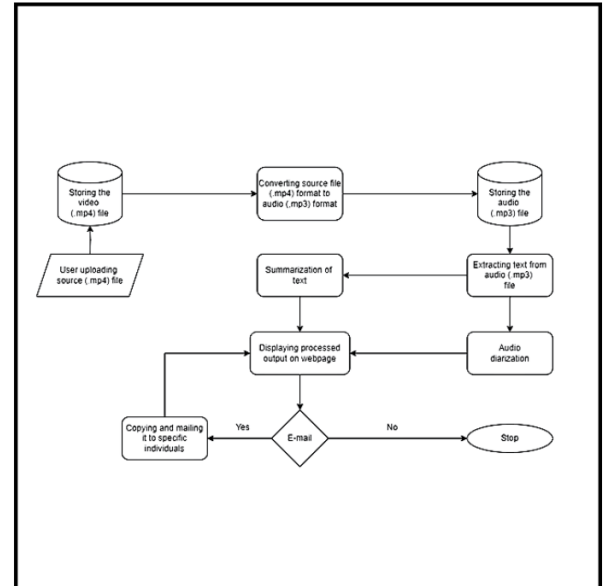


Fig. 1. Architecture of The System

II. RELATED WORK

Recent advancements in meeting summarization have been driven by the deployment of large language models (LLMs) and cutting-edge NLP techniques.

Laskar et al. [1] assessed the practical deployment of LLMs such as GPT-4 and PaLM-2 in meeting summarization, highlighting concerns over computational complexity and the need for domain-specific fine-tuning.

Asthana et al. [2] introduced a user-centric summarization system with a focus on extracting key highlights and actionable items but lacked integrated speaker attribution.

Liu et al. [3] proposed a query-driven summarization method that enhances relevance through user-specified queries but requires additional manual input. Unlike these approaches,

our system uniquely combines Gemini-based abstractive summarization with speaker diarization and automated dissemination via email, enabling a fully autonomous and context-aware summarization pipeline. Comparative experiments reveal that our system delivers concise, high-quality summaries with minimal user involvement, outperforming traditional query-based and extractive systems in both efficiency and accuracy.

Rennard et al. [4] conducted a comprehensive survey of extractive and abstractive summarization methods, covering techniques ranging from traditional rule-based systems to modern transformer-based architectures. They also explored key evaluation metrics like ROUGE and BLEU, while identifying challenges such as semantic understanding and domain adaptability.

The reviewed literature highlights significant progress in meeting summarization using both traditional NLP techniques and emerging large language models (LLMs). While extractive and abstractive approaches provide foundational methods for summarization, recent innovations such as query-focused models and action-item detection offer enhanced relevance and usability. These studies collectively emphasize the importance of accurate transcription, context-aware summarization, and speaker attribution. Building upon these insights, our work integrates robust ASR, diarization, and Gemini-based summarization to develop a practical, end-to-end meeting summarization system. By focusing on usability, automation, and accessibility, our system addresses critical gaps identified in prior research and offers a scalable solution for modern virtual collaboration.

III. METHODOLOGY

The Virtual Meeting Summarizer follows a well-structured and technically sophisticated workflow to process, analyze, and summarize meeting recordings. This systematic approach ensures accuracy, efficiency, and user accessibility.

A. Preprocessing

Accurate preprocessing is critical for enhancing both transcription fidelity and summarization quality. The uploaded audio files are initially transcribed using ASR, followed by the following preprocessing stages:

1. Noise and Filler Removal: Automated filtering of common filler words (e.g., "uh," "um") and background noise to optimize data clarity.
2. Stop Word Elimination: Removal of frequently occurring but non-informative words to emphasize critical content
3. Tokenization and Lemmatization: Sentence segmentation and word-level processing to standardize word forms and improve semantic consistency.
4. Speaker Tracking: Preservation of diarized speaker labels to ensure accurate context and attribution throughout the summarization process.

These steps are essential to ensure the summarization model focuses on structured, high-value content.

B. File Upload and Storage

The process begins with users uploading meeting recordings, typically in MP4 format, through the application's intuitive interface. These files are then securely stored in the system's local storage, setting the groundwork for subsequent processing stages. Efficient file management ensures that large datasets are handled seamlessly without performance bottlenecks.

C. Audio Extraction and Conversion

To facilitate text extraction, the system converts the uploaded MP4 file into MP3 format using the robust Python MoviePy library. This conversion is crucial as it isolates the audio stream, making it easier for further analysis and processing. MoviePy ensures a lossless and efficient transformation while maintaining the quality and integrity of the audio data.

D. Speech-to-Text Transcription

Once the audio extraction is complete, the application utilizes advanced Automatic Speech Recognition (ASR) techniques to transcribe the spoken content into readable text. This transcription process is pivotal as it transforms unstructured audio data into a structured textual representation. The system leverages AssemblyAI, a state-of-the-art ASR platform, to ensure high accuracy in speech recognition, handling variations in accents, noise, and speaker modulation effectively.

E. Summarization Using Google's Gemini-based summarization module

The transcribed text undergoes a sophisticated NLP-based summarization process. The system employs the powerful Google's Gemini-based summarization module, which applies natural language processing (NLP) and deep learning techniques to analyze, extract, and condense key discussion points. The summarization mechanism ensures that the essence of the meeting is retained while filtering out redundant information, ultimately providing users with concise, meaningful, and contextually rich summaries.

F. Speaker Diarization and Attribution

A crucial aspect of understanding virtual meetings is recognizing contributions from individual participants. The AssemblyAI diarization module is employed to distinguish and attribute speech segments to specific speakers. This process enhances clarity by associating textual content with its respective speaker, improving readability and comprehension of discussions. Speaker labels help users navigate through transcripts effortlessly, identifying who said what in a structured format.

G. System Architecture and Technology Stack

The backend of the system is built using Python, leveraging its versatility for handling audio processing, file management, and API interactions with NLP and ASR services. The frontend employs modern React.js, React Fiber, and Three.js, ensuring a dynamic, responsive, and user-friendly interface. These technologies facilitate seamless file uploads, real-time data visualization, and intuitive interactions.

H. Summary Distribution via Email

To foster seamless collaboration, the application integrates an email dissemination feature. Once summaries and diarized transcripts are generated, users can send them via email to participants who missed the meeting. The system allows users to customize messages and select recipients, ensuring that key discussions and decisions reach all relevant stakeholders. This functionality enhances accessibility and promotes inclusivity in virtual and hybrid work environments.

IV. RESULTS AND DISCUSSION

In this section, we present the implementation process of our proposed system, highlighting the key components of the user interface and demonstrating the functionality through a step-by-step workflow.

A. User Interface

Figure 2 shows the main interface of our system, designed with simplicity and user-friendliness in mind. The interface allows users to easily upload meeting recordings and view the generated summaries.

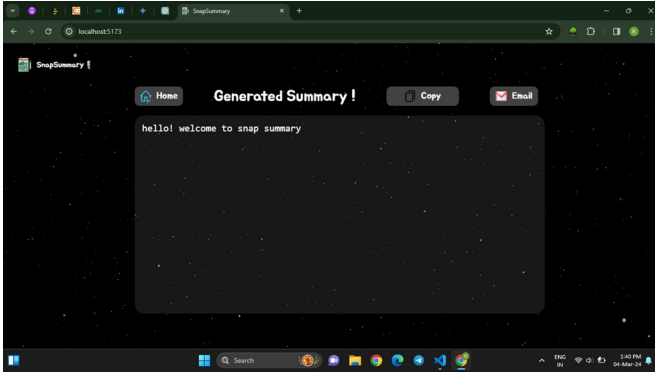


Fig. 2. User Interface of the Meeting Summarizer

B. Selecting a Meeting Recording in Video Format

The user begins by selecting a meeting recording in video format. Figure 3 shows the file selection dialog where users can browse and choose the appropriate video file.

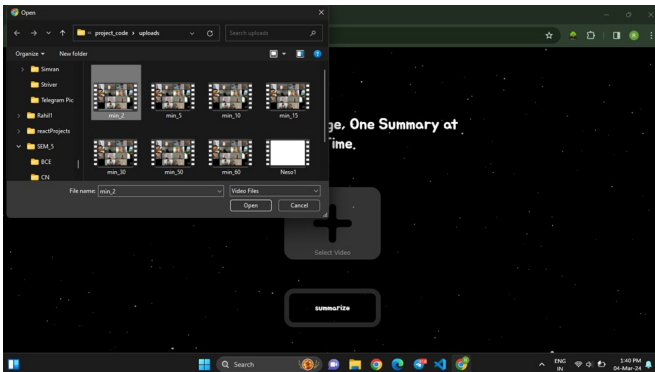


Fig. 3. Selecting the Meeting Recording File

C. Uploading the File

Once the file is selected, the user uploads it to the system as shown in Figure 4. The system then processes the file to extract audio and transcribe the meeting content.

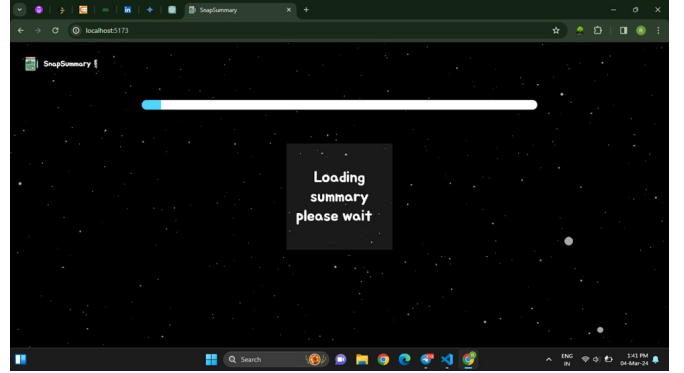


Fig. 4. Uploading the Meeting Video

D. Generating the Summary

After processing the audio, the system uses natural language processing techniques to generate a concise summary of the meeting. This is illustrated in Fig. 5.

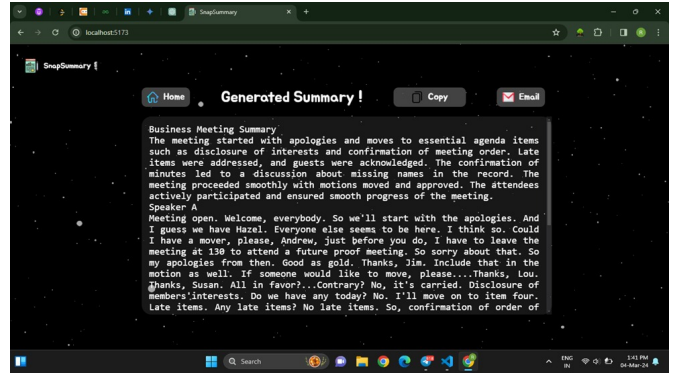


Fig. 5. System Processing and Generating Summary

E. Final Output

The final summarized text is displayed to the user, as shown in Fig. 5. This summary captures the key points of the meeting and serves as a valuable resource for future reference.

This implementation showcases the effectiveness and ease of use of the system in assisting users with automated meeting summarization.

V. CONCLUSION AND FUTURE WORK

Our study highlights the transformative potential of AI-driven summarization in enhancing virtual meeting documentation. By leveraging Natural Language Processing (NLP), speech-to-text conversion, and speaker diarization, our system effectively extracts key insights from meeting recordings, offering users concise and structured summaries. The implementation of machine learning techniques ensures that summaries

capture the essence of discussions while minimizing information overload. Additionally, the integration of email-based dissemination allows for improved collaboration, ensuring that absentees stay informed.

Despite its effectiveness, our system faces certain challenges and limitations. Accurate speaker attribution, handling overlapping speech, and improving domain adaptability remain areas that require further refinement. Additionally, while current NLP models perform well, enhancing contextual understanding and reducing hallucinations in AI-generated summaries could significantly improve reliability.

Future Research Directions

Moving forward, several areas offer opportunities for enhancing AI-powered meeting summarization:

1. Improving Summarization Quality: Future models could incorporate multimodal learning, combining audio, video, and text-based inputs to improve summary accuracy. Integrating real-time summarization features could also enhance user experience.
2. Customizable and Adaptive Summaries: Allowing users to tailor summaries based on their preferences, such as action item extraction, sentiment analysis, or topic-based filtering, could make the system more versatile.
3. Enhanced Speaker Diarization: Implementing advanced deep-learning techniques for diarization could improve speaker identification and segmentation, especially in large meetings with multiple participants and overlapping dialogues.
4. Multilingual Support: Expanding the system to support multiple languages and dialects would increase accessibility and usability in global, multilingual teams.
5. Privacy-Preserving AI Models: Research into federated learning and on-device processing could enable meeting summarization while safeguarding user data privacy and security.

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